

Advanced V&V for motor control

- Motor Control Sensitivity Analysis

Advanced V&V

- Motor control sensitivity analysis.
- Integration testing with Transient Simulation Architecture.
- Verification of algorithms that manage failure reaction.
- Sensitivity analysis on plant parameters variation.
- Automatic Execution of test.
- Robustness of controller.

User case #2

What is idea/aim behind running user case#2???

- Adding noise on V_{dc} and I_{abc} of ADC block .
- Effect of noise on Motor control sensitivity.
- To Check robustness of controller.

Types of Noise

- Power supply noise.(Analog input signal noise.)
- PCB layout
- I/O switching
- Analog input signal noise.
- Analog Source Impedance.(Resistance and Capacitance).
- Injection of leakage current.
- Electromagnetic Interference noise.
- Converter errors.(gain, linearity, offset)

Use Case #2 - Motor Control Sensitivity Analysis

Tool: Transient Simulation Architecture

Motor Control: DTC for sBPM

Stream: Windy_Mcu_sBPM_Floating

Board: WindyStripH50_230V

Motor: H27 Hefei [with fixed inductances] - Dyno Load mechanical

Description: **Evaluate motor controllability under noise /disturbance/Offset on ADC input.**

-Introduce random noise to ADC input.

Ranges:see speed\torque matrix [to be applied in dyno load mode]

Report:

- Rotor position error
- Torque estimation error
- Max input variation before failure [motor stall condition]

Use Case #2 - Motor Control Sensitivity Analysis

- Motor Inputs - Vdc and Motor currents (Iabc).

--Test to Fail Modes



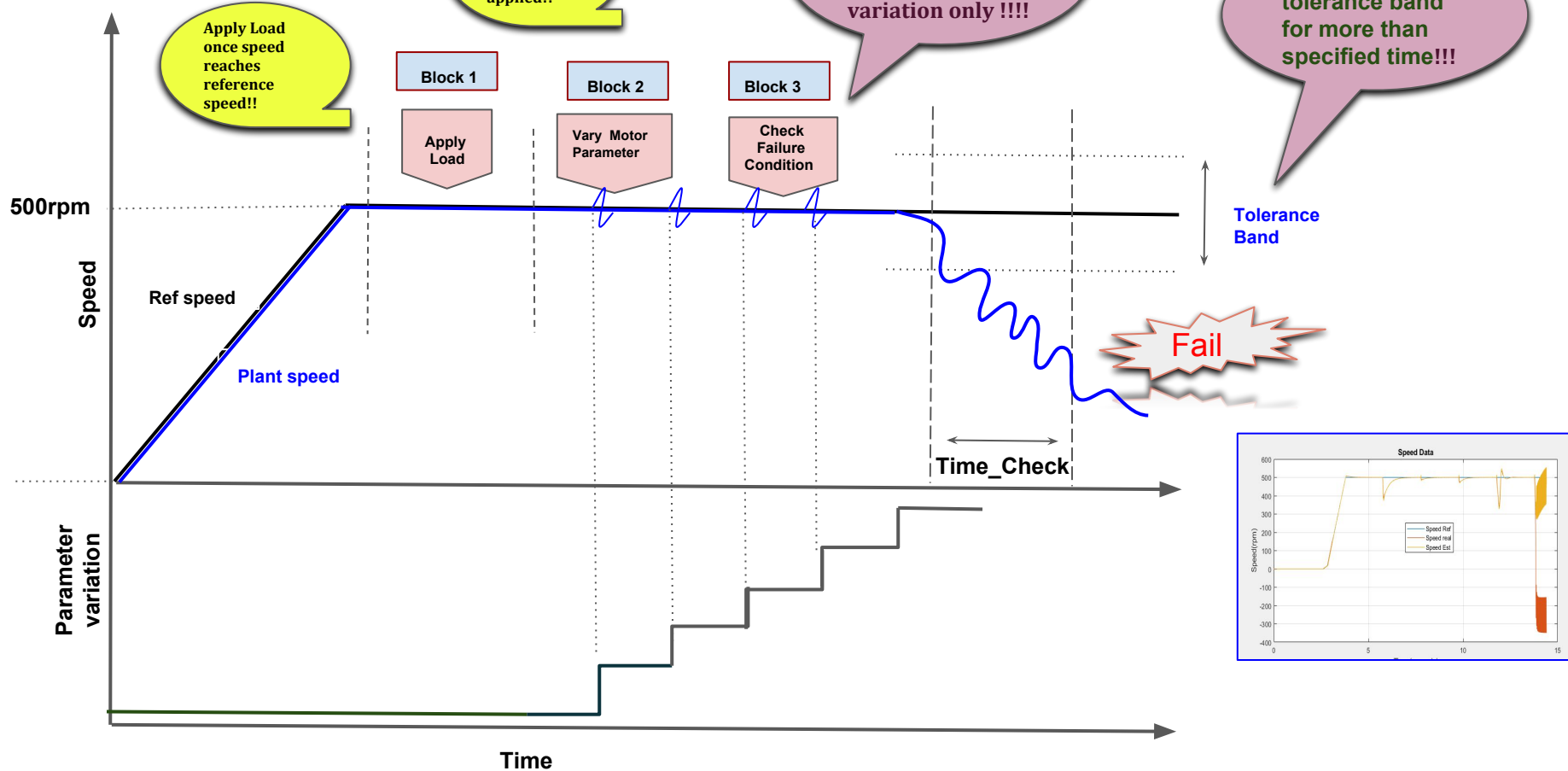
1) Add noise- Vdc

[once per time] after motor reaches steady state speed

2) Add noise-Iabc(all 3)

Speed [Drum]	Torque[Drum]		
50 rpm	0Nm	15Nm	25 Nm
500rpm	0Nm	5Nm	10Nm
1000 rpm	0Nm	3.5Nm	7Nm

Implementation



Structure

Advanced VV

Scripts

Automatic Script.m

--Allows to run the test with the given speed/torque.
--It manages the output.

Initialization.m

--Allows user to input the Parameters and limits.
--Allows the user for selection of type of load, stream and test.

Harness Model

Signal block

It Inputs the parameter variation signal to the module which is under test.

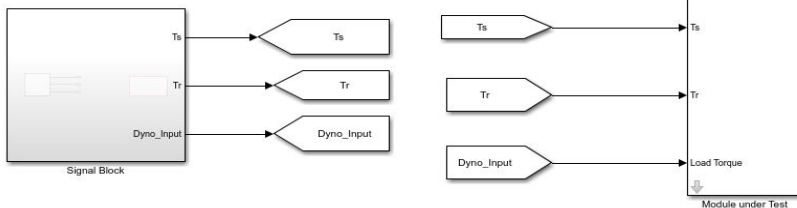
Exported block

Plant

Controller

Library

Library includes detailed documentation of the blocks created in Advanced VV.



Initialization Scripts

```
% Last Modified by v1.1 14-Feb-2019 16:08:20
%
% Author:      Madhura Patil
% Date:       October 12th, 2018
%
%
%-----
% Parameter Initialization
% Description: This Function is used to load the parameters variable given
% to simulink Model used for testing the sBPM by varying resistance and flux
% and thus interns temperature of stator and rotor temperature is varied.
%-----
%
% PLEASE NOTE -This script is used in Automatic_script_Temp.m file for calling parameters.
%-----
% Edit the below parameters to modify the response.
%-----

addpath(genpath(pwd));
%.....Input Parameter value.....%
Automatic_Test.Speed=[500;5000;1200]; % .....To enter Speed values(rpm).....
Automatic_Test.Dyno_Load_Vector=[0.5;0.8;1.3]; % .....To enter Dyno Load values(Nm).....
Automatic_Test.Acceleration=500; % .....To enter Acceleration of Speed(m/s)....
Automatic_Test.Speed_Tolerance=20; % .....To enter Speed Tolerance in(%).....
Automatic_Test.Percentage_Variation=50; % .....To enter the Percentage Variation of U

%...Flag setting for selection of variation of parameter.....%

%Uncomment either one of it as per the selection of test,that is by varying

%Automatic_Test.parameter_var= 'Flux';
Automatic_Test.parameter_var= 'Rs';

%.....Demagnetisation value.....%
Automatic_Test.dm=-0.0021;

%.....Input Parameter value to Matlab Functions.....%

%.....Status Function.....%
Automatic_Test.Status_Time_Check=2; %...Status time check in (sec)
Automatic_Test.Status_Tolerance=0.05; %...Status Tolerance time in (s)

%.....Dyno Input Function.....%
Automatic_Test.Dyno_Step_Size=0.001; %...Dyno Step Size value in (s)

%.....Parameter Input Function.....%
Automatic_Test.Para_time_check=2; %...Generation of Parameter in (s)
```



Instruction to use m file



Enter/modify the motor parameters such as speed acc load etc



Enter/modify the controller parameters such as speed tolerance,% variation etc



Selection of test. For e.g by imposing noise on Vdc/labdc.



Enter/modify other parameters as time checks for step size,failure time etc.

Automatic Scripts

To collect values of varied motor parameters at each step interval.

To plot the Report.

Naming the figure against its speed and torque.

Plot and format each figure by giving label,title,legends etc.

To save the Report and .fig file in created folder.

Function to calculate statistics (mean and standard deviation) of each signal at each interval.

Instructions

Open harness model and call initialization file.

Create folder to save figure and .csv file.

Masking parameters as per selection of file.

Create a Report format.

Create loop to run the test against Speed and Torque/Parameter matrix.

Simulate the Harness Model.

To collect statistics such as mean/standard deviation of Theta and Torque error.

```
%.....To collect the parameters values.....%
Speed_ref_prm=Automatic_Test.Speed_Ref*ones(size(Theta_Error_mean_values)); %..
Dyno_ref_prm = Automatic_Test.Dyno_Load * ones(size(Theta_Error_mean_values)); %..

%.....To get the data of parameter variation....%
if (strcmp(Automatic_Test.parameter_var,'Rs'))
step = Automatic_Test.Percentage_Variation*Motor.Electrical.Rs_a_Ref/100;
Parameter_Var = (Motor.Electrical.Rs_a_Ref:step:(Motor.Electrical.Rs_a_Ref+step*(1e
elseif(strcmp(Automatic_Test.parameter_var,'Flux'))

step = Automatic_Test.Percentage_Variation*Motor.Electrical.PM_Flux_f01/100;
Parameter_Var = (Motor.Electrical.PM_Flux_f01:step:(Motor.Electrical.PM_Flux_f01+st
end

%.....To update the results table.....%
Results = [Results ; table(Speed_ref_prm',Dyno_ref_prm',Parameter_Va
Theta_Error_std_values',Torque_Error_mean_values',Torque_Error
Speed_Plant_mean_values',Speed_Plant_std_values', ...
'VariableName',{'Speed_ref','Torque_ref', Automatic_Test.parameter
'Theta_error_std', 'Torque_error_mean', 'Torque_error_std', 'Sp

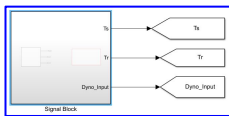
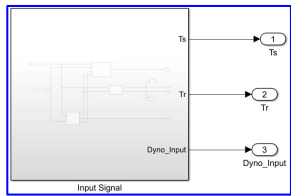
%.....Naming the figure file against speed and torque....%
r=strcat('Speed_',num2str(Automatic_Test.Speed(i)),'_Load_',strrep(n

%..... To Plot the Speed, torque and theta error.....%
y = figure('Name','Run','NumberTitle','off');
subplot(2,2,1);
plot(Speed);
ylabel('Speed(rpm)')
title('Speed Data')
legend('Speed Ref','Speed real','Speed Est')
grid on;
subplot(2,2,2);

function [interval_index] = get_intervals(parameter_signal)
parameter_signal_shifted = [parameter_signal(1) parameter_signal];
parameter_signal = [parameter_signal parameter_signal(end)];
interval_index = parameter_signal_shifted - parameter_signal;

k=find(interval_index==0);
if k
%....Adding first interval with nominal value....%
interval_index = [k(1)-(k(2)-k(1)) k];
else
interval_index = [];
end

function [mean_values,std_values] = statistics(signal, interval_index)
n=length(interval_index)-1;
mean_values = zeros(1,n);
std_values = zeros(1,n);
for i=1:n
interval=interval_index(i+1)-interval_index(i); % To get the interv
start_1=interval_index(i)+floor(interval*0.2); % To get the data st
final=interval_index(i+1)-floor(interval*0.01); % To get the data b
windowed_value=signal(start_1:final);
mean_values(i)=mean(windowed_value);
std_values(i)=std(windowed_value);
end
end
```

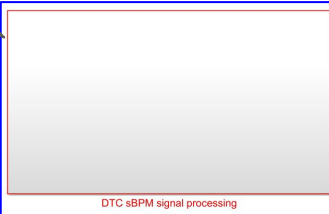


Input Signal Block

Signals Input

Signals and Scope

Signal Processing scopes



Block-1
Speed
Dyno_Load

Block-2
Parameter
Variation

Block-3
Condition
check

Speed
Accele

Speed Input

Speed attained

Dyno Status

Dyno input

Dyno Input

Dyno Load Input

Step Signal

Noise Input

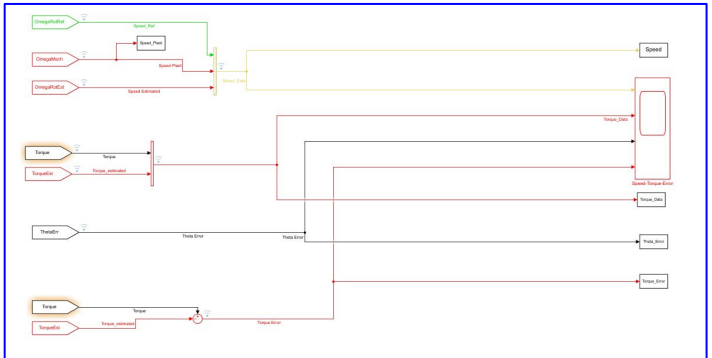
Vdc labc

Dyno status
Speed

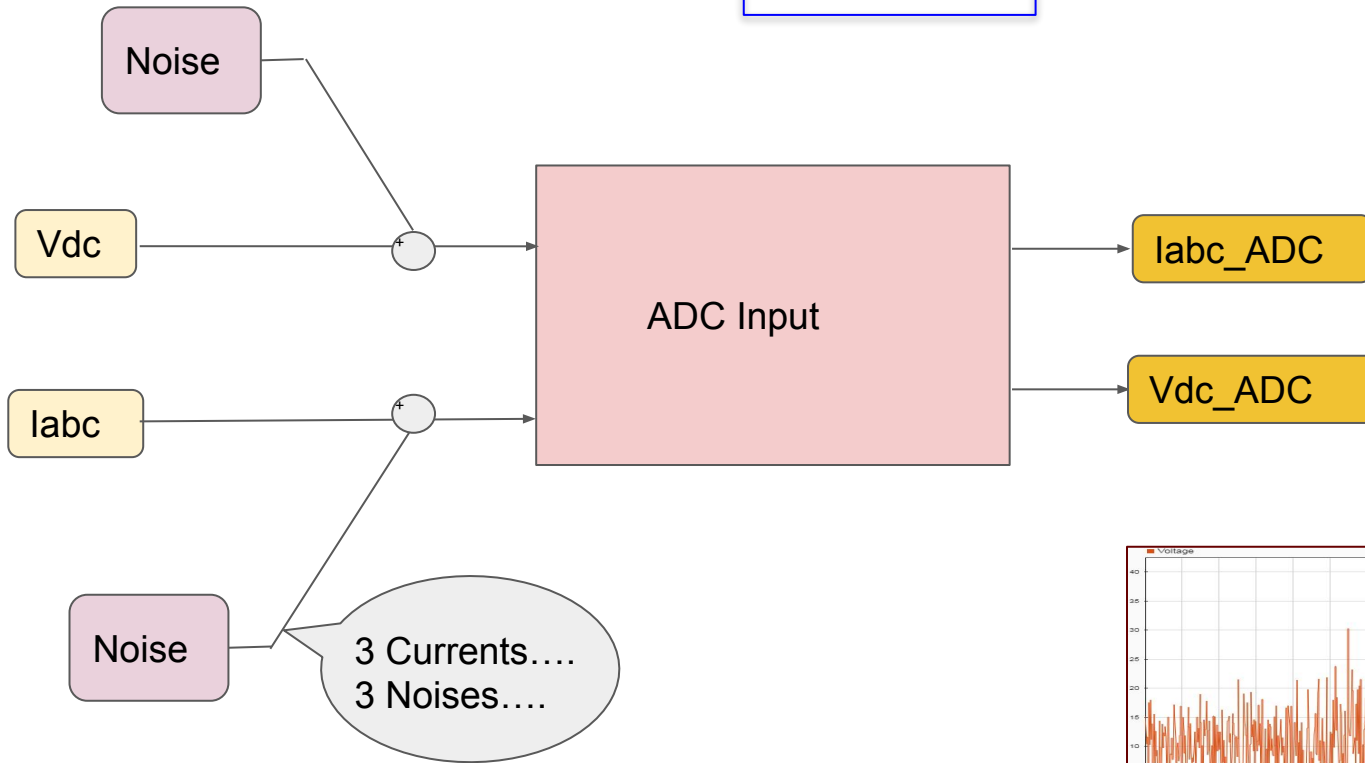
Condition Check

If speed is out of tolerance bound for more than specified time

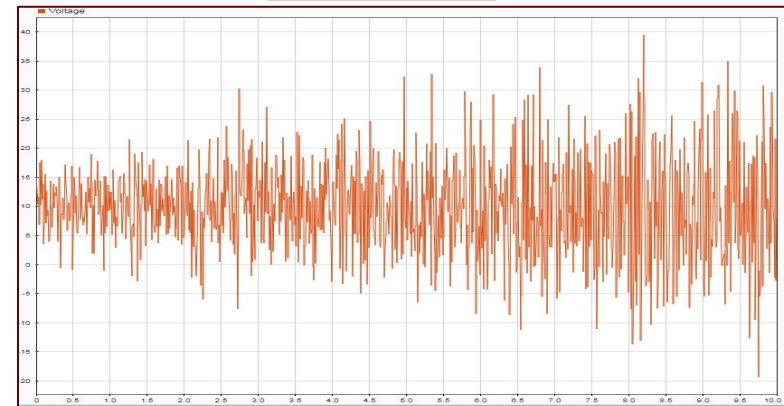
Stop



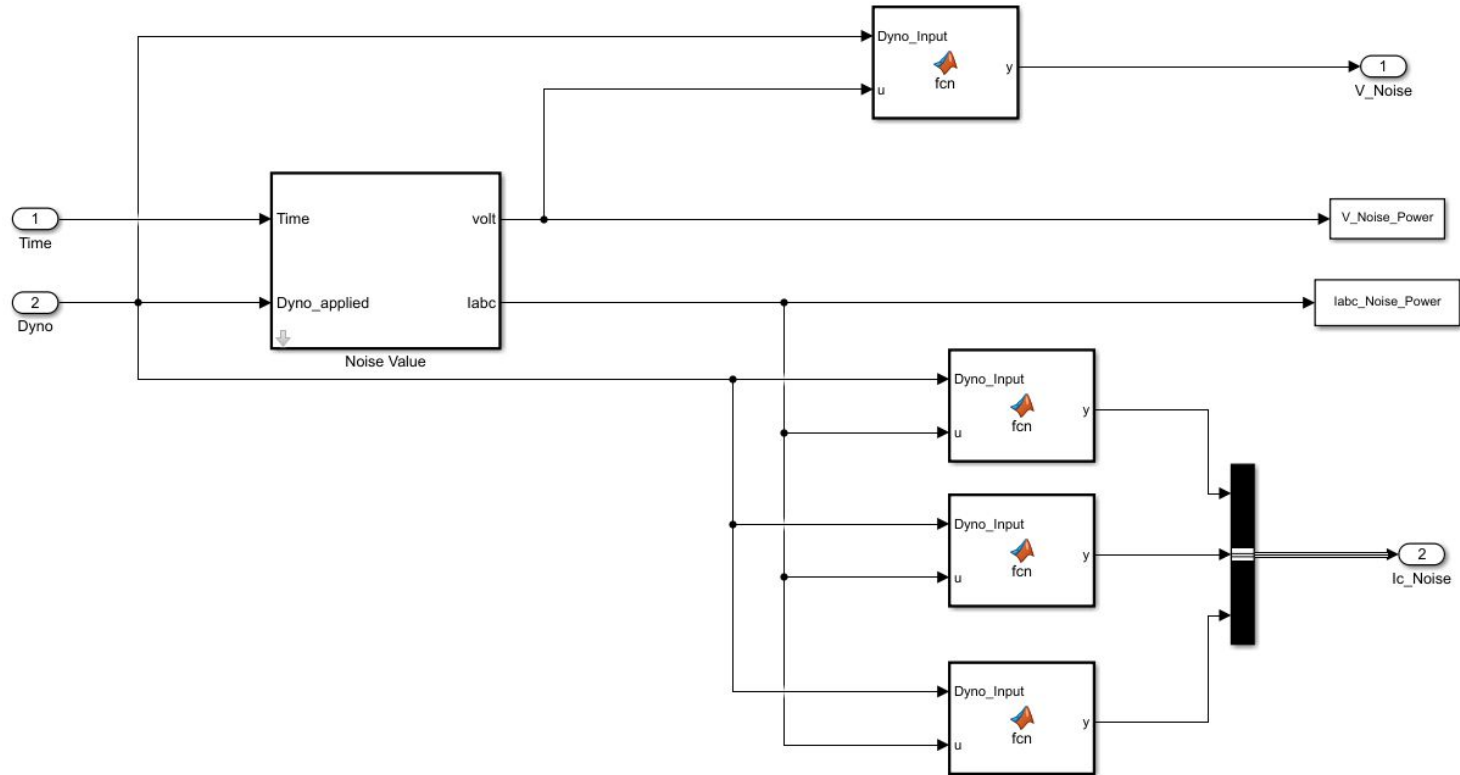
ADC Block



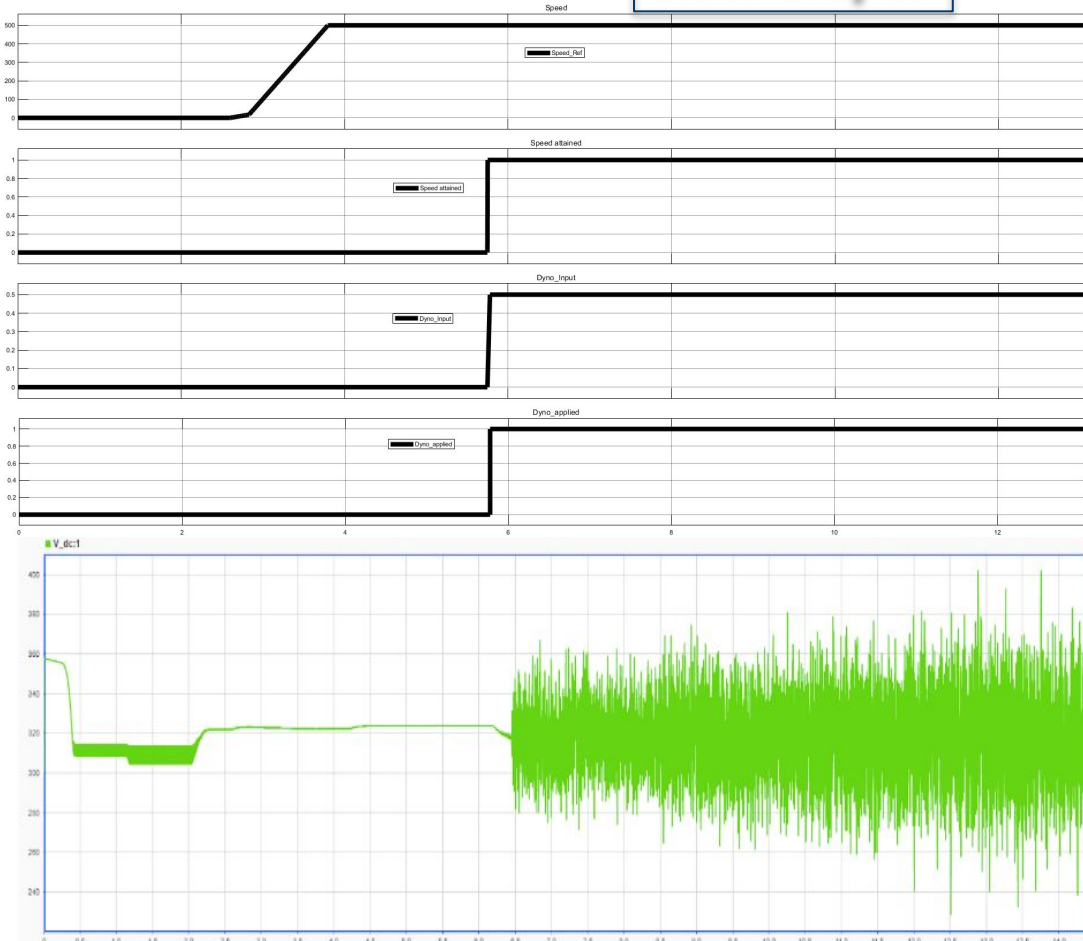
Random Noise



Noise Block



Sinal Graphs



Speed Graph



When plant speed reaches ref speed, **Speed attained** Status is high.



Speed attained Status is high **Dyno load** has applied.

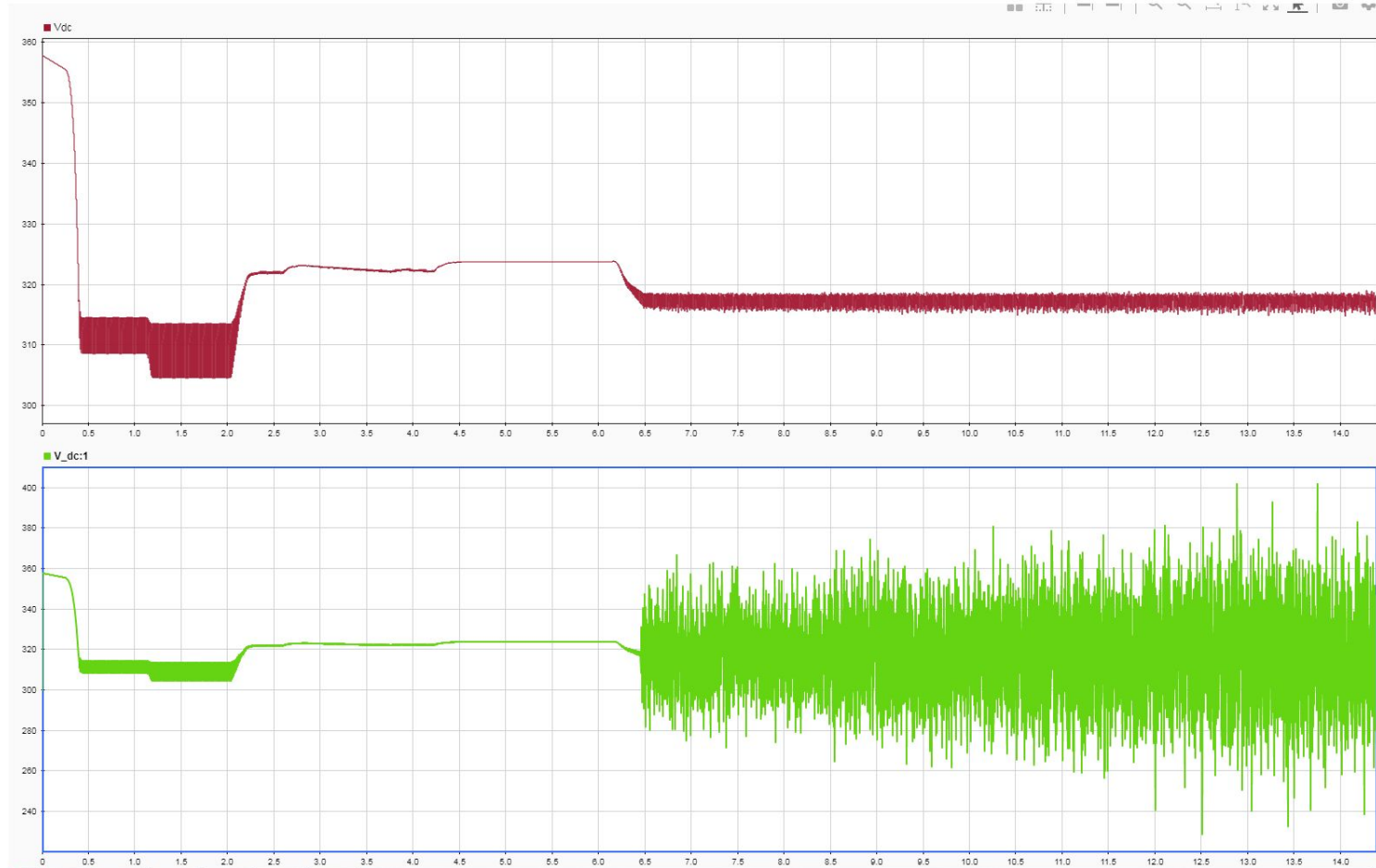


When Dyno load has applied, **Dyno applied** status is high.



Once Dyno applied status is high, start imposing noise.

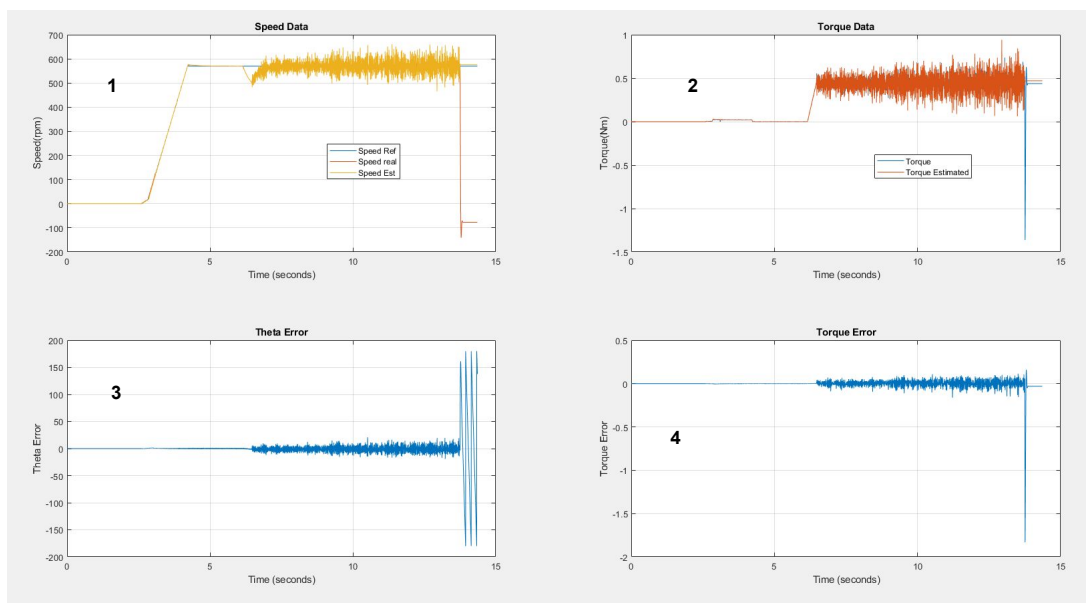
Vdc Vs Vdc_Noise



Results

Format of figure

1. Speed-Reference,Plant and Controller speed
2. Torque- Plant and Controller torque.
3. Theta Error
4. Torque error



Format of report

[illegible]

Folder structure

MCU_Sim_Arch

advanced_V_V

lib (Library)

This folder contains library of each matlab blocks created for advanced VV testing purpose.

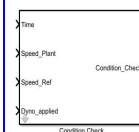


Go to library browser

Condition Check_Temp

Description

This Matlab function block is used to keep check on failure of the test.



Input

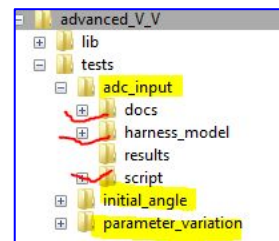
1. Time is real simulation time which is used to calculate failure conditions.
2. Speed_Plant is the output speed from the plant.
3. Speed_Ref is the given reference speed by the user.
4. Dyno_Applied is the status to know whether Dyno load is applied or not.

Output

The output Condition Check will send the signal to stop the simulation if failure condition is true

tests

--adc_input
--initial-Angle
--Parameter
variation



Docs-Help document/guidelines to conduct tes.

Harness_model- Contains harness model as .slx file.

Results - Folder to save the results.

Script - Contains two script, Automatic and Initialization.m files